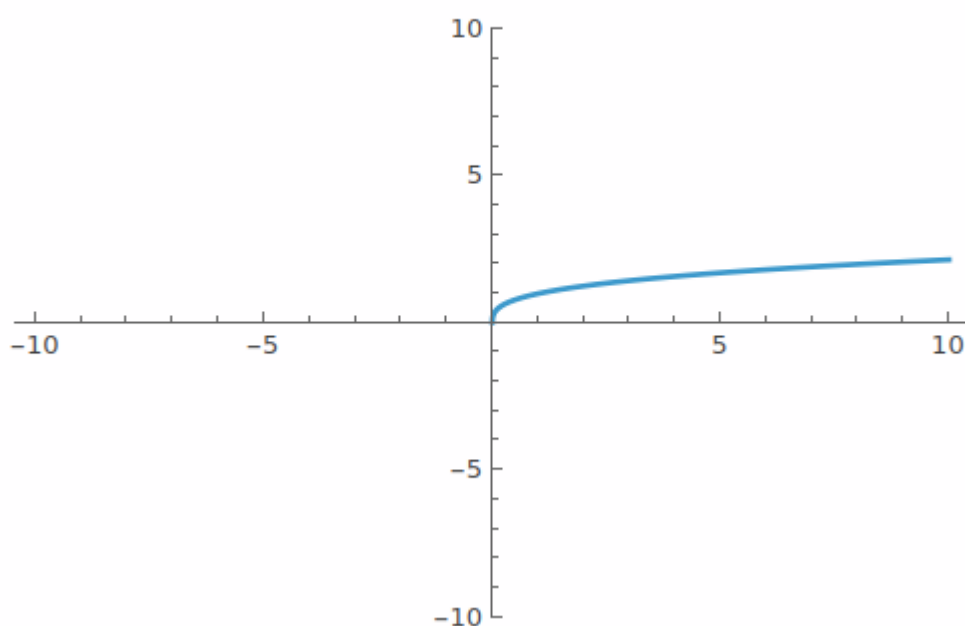


Some Mathematica things to know

When to use `Surd[...]`

- Mathematica doesn't like having negative values raised to fractional powers when rendering
 - Usually not a problem for algebraic manipulation
 - IS a problem when visualising
- This means you won't see certain parts (usually negative) of fractional power graphs
 - Eg: Inverse of $f(x) = x^3$

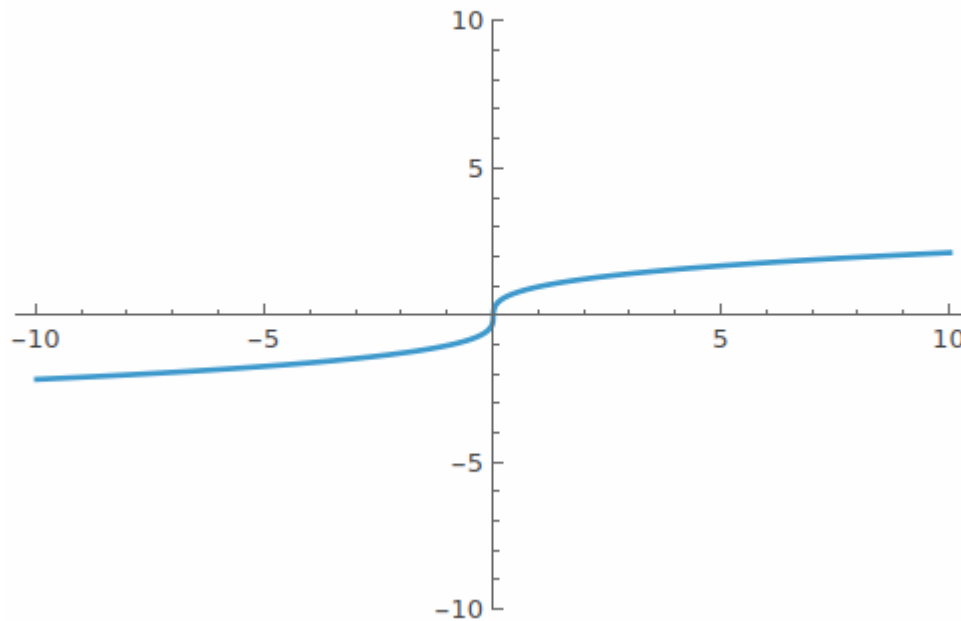
```
Plot[x1/3, {x, -10, 10}, PlotRange -> 10]
```



- Where did the negative part of the graph go?

- Need to use `Surd[x, 3]` instead.

`Plot[Surd[x, 3], {x, -10, 10}, PlotRange -> 10]`



- Alternative: can also use `CubeRoot[x]` - Although I don't recommend this, just go for `Surd[x, n]` to get $x^{1/n}$ for any n

Difference between SolveAlways and FindInstance

- `SolveAlways` : Finds values of unknown parameters that makes an equation of x, y true for all values of x, y
 - This treats x, y as variables of a function, whereas the parameters are fixed
 - Takes in the function variables (variables that are free)
- `FindInstance` : Finds a set of values for unknown variables that makes the equation true
 - This doesn't distinguish between the types of variables
 - You are not letting any of the variables be free. All are considered fixed
 - Takes in all unknown variables
- $f(x) = ax^2$, x is a function variable and a is a parameter, they are treated
 - differently in `SolveAlways`, same in `FindInstance`

EXAMPLES

- Q: "Find values of a, b, c such that $x^2 - 3x + by + a = (x - b)(x - c) + cy \quad \forall x, y \in \mathbb{R}$ "
 - Finding values of unknown PARAMETERS so that the equation holds. x, y are function variables that are free (can be anything)

```
SolveAlways[x^2 - 3 x + b y + a == (x - b) (x - c) + c y, {x,y}]
> {{a -> 9/4, b -> 3/2, c -> 3/2}}
```

2. Q: "Find values of $a, b, c \in \mathbb{R}$ such that $\frac{a+b-c}{3} - 2 = \frac{a}{c}(b-10) + 1$ "

- Simply just finding values of all unknowns so that the equation holds. None of them are function variables.

```
FindInstance[(a + b - c)/3 - 2 == a/c (b - 10) + 1, {a, b, c}]
```

```
> {{a -> -(7/26), b -> 1, c -> -1}}
```